MODULE 2: FOUNDATIONS OF ELECTRONICS AND COMMUNICATION SYSTEMS

(CO: 2)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACEE101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 8 Topics): (1) PN Junction Diode: Symbol, circuit diagram, characteristics, single-phase full-wave diode bridge rectifier, (2) Bipolar Junction Transistor (BJT): Symbol, circuit diagram, characteristics (CE configuration), (3) MOSFET: Symbol, circuit diagram, characteristics (N-Channel Enhancement MOSFET), (4) Introduction to Operational Amplifiers: Symbol, Inverting, Non-inverting, Voltage Follower, Adder, (5) Electromagnetic Spectrum & Elements of Communication Systems, (6) Overview of cellular communication, (7) Fundamentals of Satellite Communication and Radar, and (8) Contemporary Issues.

1 VIT FAT Exam Blueprint & Master Formula Sheet

Module 2 carries **20 to 25 marks** in BACEE101 FAT:

1. **Question Type A (10 Marks):** Diode Bridge Rectifier: Full circuit schematic, operation during positive and negative half-cycles, derivation of DC voltage ($V_{\text{dc}} = \frac{2V_m}{\pi}$), ripple factor ($\gamma = 0.482$), and efficiency ($\eta = 81.2\%$).
2. **Question Type B (10 Marks):** Operational Amplifier circuits: Derivation of closed-loop voltage gain for Inverting amplifier ($A_v = -\frac{R_f}{R_1}$), Non-Inverting amplifier ($A_v = 1 + \frac{R_f}{R_1}$), Summing amplifier, and Voltage follower using the virtual short-circuit concept.
3. **Question Type C (10 Marks):** Communication Systems & Radar: Elements of communication block diagram, cellular frequency reuse, or Radar Range Equation derivation ($R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{\min}}\right]^{\frac{1}{4}}$).

Device / System	Governing Equation / Formula	VIT Exam Key Insight
Diode Bridge Rectifier	$V_{\text{dc}} = \frac{2V_m}{\pi} \approx 0.636 V_m, \quad \text{PIV} = V_m$	Ripple frequency is $2f = 100$ Hz (for 50 Hz AC mains); ripple factor $\gamma = 0.482$.
BJT Current Relation	$I_E = I_B + I_C, \quad I_C = \beta I_B + I_{\text{CEO}}, \quad \beta = \frac{\alpha}{1-\alpha}$	In CE active region, base-emitter is forward-biased, collector-base is reverse-biased.
MOSFET Drain Current	$I_D = \frac{1}{2}\mu_n C_{\text{ox}} \left(\frac{W}{L}\right)(V_{\text{GS}} - V_{\text{th}})^2$	Saturation occurs when $V_{\text{DS}} \geq V_{\text{GS}} - V_{\text{th}}$; pinch-off condition.
Inverting Op-Amp Gain	$V_{\text{out}} = -\left(\frac{R_f}{R_1}\right) V_{\text{in}}$	Virtual ground at inverting terminal: $V_- = V_+ = 0$ V.
Non-Inverting Op-Amp Gain	$V_{\text{out}} = \left(1 + \frac{R_f}{R_1}\right) V_{\text{in}}$	Input impedance is extremely high ($> 10^{12}\Omega$), zero input bias current.
Cellular Capacity	$C = N \times K, \quad \frac{S}{I} = \frac{(\sqrt{3N})^{\gamma}}{i_0}$	Frequency reuse cluster size $N = i^2 + ij + j^2$ ($N = 3, 4, 7, 12$).
Radar Range Equation	$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{\min}}\right]^{\frac{1}{4}}$	Received echo power falls off inversely with the fourth power of target distance R^4.

2 Semiconductor Devices & Circuits

2.1 Full-Wave Diode Bridge Rectifier

A bridge rectifier uses four diodes (D_1, D_2, D_3, D_4) connected in a closed bridge loop across AC secondary voltage $v_{s(t)} = V_m \sin(\omega t)$:

- **Positive Half Cycle ($0 < \omega t < \pi$):** Terminal A is positive, B is negative. Diodes D_1 and D_2 are forward-biased, conducting current through load R_L from top to bottom. Diodes D_3 and D_4 are reverse-biased.
- **Negative Half Cycle ($\pi < \omega t < 2\pi$):** Terminal B is positive, A is negative. Diodes D_3 and D_4 are forward-biased, directing current through load R_L in the **exact same direction**.
- **Average DC Voltage:**

$$V_{\text{dc}} = \frac{1}{\pi}\int_0^{\pi} V_m \sin(\omega t) d(\omega t) = \frac{2V_m}{\pi} \approx 0.636 V_m$$

- **RMS Voltage:** $V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \approx 0.707 V_m$.
- **Ripple Factor:** $\gamma = \sqrt{\left(\frac{V_{\text{rms}}}{V_{\text{dc}}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = \sqrt{1.2337 - 1} = 0.482$ (48.2%).
- **Peak Inverse Voltage (PIV):** Maximum reverse voltage across non-conducting diodes is V_m (unlike center-tapped full-wave which requires $2V_m$).

2.2 Operational Amplifiers (Op-Amps)

An ideal operational amplifier possesses:

1. Infinite open-loop voltage gain ($A_{\text{OL}} \to \infty$).
2. Infinite input impedance ($R_{\text{in}} \to \infty \Rightarrow I_+ = I_- = 0$).
3. Zero output impedance ($R_{\text{out}} = 0$).
4. Infinite bandwidth and zero input offset voltage.

Golden Rules of Op-Amps:

- **Rule 1 (Virtual Short):** With negative feedback, the op-amp output drives the inverting input voltage to equal the non-inverting input voltage: $V_- = V_+$.
- **Rule 2 (Zero Input Current):** Due to infinite input impedance, no current enters or leaves the input terminals: $I_+ = I_- = 0$.

1. Inverting Amplifier: Connect non-inverting terminal to ground ($V_+ = 0$ V). By Rule 1, inverting terminal is at virtual ground ($V_- = 0$ V). Applying KCL at the V_- node:

$$\frac{V_{\text{in}} - 0}{R_1} + \frac{V_{\text{out}} - 0}{R_f} = 0 \Rightarrow \frac{V_{\text{out}}}{R_f} = -\frac{V_{\text{in}}}{R_1} \Rightarrow A_v = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_f}{R_1}$$

2. Non-Inverting Amplifier: Input signal V_{in} is applied directly to V_+, so $V_+ = V_{\text{in}}$. By Rule 1, $V_- = V_{\text{in}}$. Applying KCL at the V_- node:

$$\frac{0 - V_{\text{in}}}{R_1} + \frac{V_{\text{out}} - V_{\text{in}}}{R_f} = 0 \Rightarrow \frac{V_{\text{out}}}{R_f} = V_{\text{in}}\left(\frac{1}{R_1} + \frac{1}{R_f}\right) \Rightarrow V_{\text{out}} = \left(1 + \frac{R_f}{R_1}\right)V_{\text{in}}$$

3. Voltage Follower (Buffer): Setting $R_f = 0$ and $R_1 \to \infty$ yields $V_{\text{out}} = V_{\text{in}}$, providing unity gain ($A_v = 1$), infinite input resistance, and zero output resistance to prevent loading effects.

3 Communication Systems, Cellular & Radar

3.1 Elements of Communication Systems

Every electronic communication system contains:

1. **Information Source & Input Transducer:** Converts speech, video, or data into baseband electrical signals.
2. **Transmitter:** Modulates the baseband signal onto a high-frequency carrier wave for efficient radiation.
3. **Transmission Channel:** Physical medium (guided coax/fiber or unguided free-space wireless) that attenuates signal and introduces additive noise.
4. **Receiver:** Amplifies, filters, and demodulates the carrier to recover the original baseband signal.

3.2 Cellular Concepts & Frequency Reuse

To serve millions of wireless subscribers within limited frequency spectrum:

- Geographic coverage is partitioned into hexagonal cells, each served by a base transceiver station (BTS).
- The available frequency band is divided into N channel groups and assigned to adjacent cells forming a **cluster**.
- The exact frequency set is reused in cells located at distance $D = R\sqrt{3N}$, where R is cell radius and N is cluster size ($N = 3, 4, 7, 12$).

3.3 Radar Principles & Radar Range Equation

RAdio Detection And Ranging (RADAR) transmits short microwave pulses and detects reflections from target objects:

$$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 S_{\min}} \right]^{\frac{1}{4}}$$

where P_t is transmitter peak power, G is antenna directive gain, λ is wavelength, σ is target radar cross section, and $S_{\min}$ is minimum detectable receiver signal.